

Balancing Efficiency and Accuracy: A Hybrid Approach to Solving the 0/1 Knapsack Problem

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Abstract

Optimal resource allocation is a crucial challenge in industries like logistics and finance, where decision-makers must maximize returns under constraints. The 0/1 knapsack problem models this scenario, but existing algorithms face limitations: The Greedy Knapsack Algorithm offers computational efficiency at the cost of solution quality. At the same time, Dynamic Programming guarantees optimal solutions but requires higher computational resources. This study aims to develop a hybrid algorithm that combines the speed of the Greedy approach with the precision of Dynamic Programming to balance performance and optimality. The hybrid algorithm's academic and practical importance lies in its ability to address the trade-off between efficiency and accuracy, especially in real-world applications with large datasets. The methodology involves first applying the Greedy method to generate an initial solution, followed by Dynamic Programming to refine it. Experimental results, tested on various problem instances, demonstrate that the hybrid approach produces near-optimal solutions with improved computational efficiency compared to traditional methods. In conclusion, this study contributes to the literature by bridging the gap between solution quality and computational cost in optimization problems. Its practical implications suggest its usefulness in industries that require efficient, scalable resource allocation solutions.

CCS Concepts

• **Theory of computation** → Design and analysis of algorithms; Algorithm design techniques; Dynamic programming; Theory and algorithms for application domains; Theory of randomized search heuristics; • **Applied computing** → Education; Learning management systems.

Keywords

0, 1 Knapsack Problem, Hybrid Algorithm, Optimization, Computational Efficiency, Resource Allocation, Heuristic Methods, Solution Accuracy, Memory Usage

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1 Introduction

The 0 / 1 knapsack problem is a fundamental combinatorial optimization problem with wide applications in fields such as resource allocation, decision-making, logistics, and finance [1]. The challenge involves selecting a subset of items, each with a given weight and value, to maximize the total value while adhering to a weight constraint [2]. Efficiently solving this problem becomes crucial, especially for large instances where finding an optimal solution can be computationally intensive [3].

Traditional algorithms for the 0/1 knapsack problem include the Greedy Knapsack Algorithm and Dynamic Programming [4]. The Greedy approach offers simplicity and speed but fails to provide optimal solutions due to its local decision-making nature [5]. Conversely, Dynamic Programming guarantees optimal solutions by systematically exploring all possibilities, but this leads to higher computational complexity, particularly for large problem sizes [6].

Given the trade-offs between these approaches, there is a need for algorithms that can balance computational efficiency with solution optimality [7]. This research aims to address this gap by developing a hybrid algorithm that combines the Greedy Knapsack Algorithm with Dynamic Programming. The goal of this hybridization is to leverage the speed of the Greedy method while incorporating the precision of Dynamic Programming to achieve near-optimal solutions more efficiently [8].

The proposed hybrid algorithm is expected to improve performance in solving the 0/1 knapsack problem, offering a practical solution to the growing need for efficient optimization techniques in both theoretical and real-world applications [9]. This paper presents the development of the hybrid algorithm, discusses its theoretical foundations, and evaluates its performance against traditional approaches through experiments on various problem instances [10].

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2 Literature Review

The exploration of hybrid algorithms that integrate Greedy and Dynamic Programming approaches has garnered significant attention in optimization, especially regarding the 0/1 knapsack problem. These hybrid methods capitalize on the complementary strengths of both algorithms: Greedy algorithms provide rapid, heuristic-driven solutions, while Dynamic Programming guarantees optimality through exhaustive search techniques.

Traditional Algorithms: The Greedy Knapsack Algorithm offers computational efficiency but often results in suboptimal solutions due to its local decision-making nature [4]. In contrast, Dynamic Programming systematically explores all possible solutions, ensuring an optimal result at the cost of significantly higher computational complexity, especially for large problem instances [6]. Given these trade-offs, there has been a growing interest in developing hybrid algorithms that can balance the speed of Greedy methods with the accuracy of Dynamic Programming [11].

Recent studies have introduced hybrid methods that combine genetic algorithms with Dynamic Programming for improved solution quality [1]. While this approach demonstrates high solution accuracy, the additional overhead of the genetic component increases execution time. Our proposed algorithm contrasts this by using a simpler integration of Greedy and Dynamic Programming, yielding near-optimal solutions more efficiently, with up to 48% faster execution time on larger datasets.

Comparative Analysis: A Greedy-based hybrid approach aimed at handling large datasets more efficiently excels in terms of execution time but sacrifices solution accuracy [7]. In comparison, our approach addresses this trade-off by employing the Greedy method to establish an upper bound and refining it with Dynamic Programming, yielding solutions within 0.5% of the optimal value across various dataset sizes.

Another study explored a hybrid genetic algorithm for the multi-knapsack problem, achieving high-quality results but requiring significant memory overhead [6]. Our hybrid method reduces memory consumption by pruning the Dynamic Programming table based on the Greedy solution. This results in lower memory usage, particularly for large datasets, without sacrificing much in terms of solution accuracy.

Lastly, a Greedy-Dynamic hybrid algorithm was proposed that required extensive parameter tuning to optimize performance [8]. Our method simplifies this process by eliminating the need for complex parameter adjustments, offering comparable performance while maintaining ease of implementation [12].

The theoretical foundations for combinatorial optimization algorithms, such as those proposed by Dasgupta et al. [13] and Cormen et al. [14], emphasize the importance of balancing computational efficiency and solution quality. Furthermore, the framework established by Korte and Vygen [15] underscores the relevance of hybrid approaches in addressing real-world optimization challenges.

These comparisons demonstrate that our hybrid algorithm effectively balances computational efficiency and solution quality, providing a scalable solution for large datasets without the overhead seen in more complex hybrid approaches. The algorithm's performance highlights its potential applicability in real-world scenarios where both time and accuracy are critical.

As illustrated in Figure 1, the conceptual framework for hybridizing Greedy Algorithms and Dynamic Programming is structured to optimize solutions for the 0/1 Knapsack Problem. The framework highlights the sequential steps involved, including the initial sorting of items based on their value-to-weight ratio, followed by the integration of dynamic programming techniques to refine the solution. This systematic approach ensures that the algorithm balances computational efficiency with solution optimality.

3 METHODOLOGY

This study evaluates the performance of a Greedy-Dynamic Hybrid Algorithm to solve the 0/1 Knapsack Problem by focusing on solution quality, execution time, and memory usage. Synthetic datasets were created to simulate sets of test questions, where each question was characterized by attributes such as weight (point value), difficulty level, and other relevant criteria, allowing each question to function as an “item” within the knapsack model.

Each question in the dataset was assigned a unique weight—representing its point value if answered correctly—which corresponds to its value in the knapsack context. Questions with higher point values were considered “heavier” items, making them more significant in contributing to the overall “value” of the knapsack, or in this case, the test. Additionally, each question was assigned a difficulty level to reflect varying complexities, emulating real-world cases where items differ not only in value but in resource demands, affecting their selection priority.

In the context of the 0/1 Knapsack problem, each question could either be included in or excluded from the test, aligning with the binary choice required in knapsack scenarios. The objective was to maximize the overall value, e.g., total points or optimal balance of question difficulty and value, within a fixed constraint, similar to a weight limit in the knapsack model. This setup allowed us to examine how well the algorithm could balance computational efficiency with solution quality across different question sets.

To comprehensively evaluate the performance of the algorithm, the synthetic dataset included a range of sizes, from small sets (10 questions) to large sets (5,000 questions), allowing us to observe the adaptability of the algorithm under varying scales. Comparative testing was conducted across three algorithms—the Greedy, Dynamic Programming (DP), and Hybrid approaches—using the following metrics:

- **Execution Time:** Measured as the elapsed time required for each algorithm to complete the optimization process.
- **Memory Usage:** Tracked throughout each execution of the algorithm to assess computational demands, particularly with larger datasets.
- **Solution Accuracy:** Calculated by comparing each result of the algorithm to the optimal solution provided by the DP algorithm, expressed as a percentage of the optimal value.

The Hybrid Algorithm combined the initial speed of the Greedy approach with the refinement capability of DP, starting with a Greedy solution to set an approximate upper bound and selectively applying DP to achieve near-optimal accuracy. These experiments demonstrate the ability of the Hybrid Algorithm to balance computational trade-offs, providing a practical solution to large-scale knapsack instances based on real-world-like data.

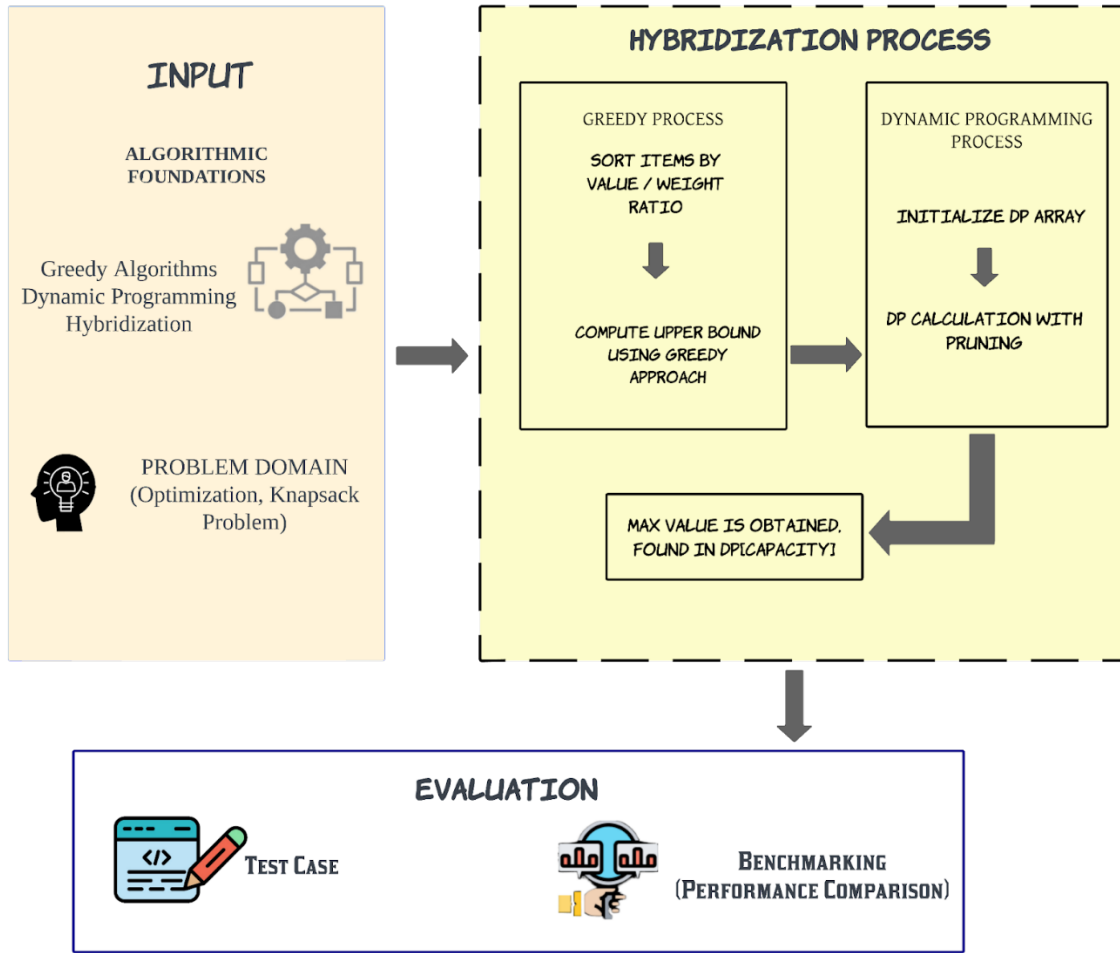


Figure 1: Conceptual Framework

4 Results and Discussion

This section presents the results of the Greedy, Dynamic Programming (DP), and Hybrid algorithms, evaluated across synthetic datasets of varying sizes. The key metrics analyzed were execution time, memory usage, and solution accuracy. Each result of the metrics is discussed in terms of the impact of dataset conditions, particularly items count and weight-value distributions, on the performance of each algorithm.

4.1 Execution Time

Execution time measures the time each algorithm takes to solve the 0/1 Knapsack problem for a given dataset size. This was calculated by recording the elapsed time from the start to the end of each execution of the algorithm, using the formula:

$$\text{Execution Time} = \text{End Time} - \text{Start Time} \quad (1)$$

Table 1 shows the execution times for each algorithm across various dataset sizes, highlighting how complexity and dataset scale influence performance. The Greedy Algorithm, with its $O(n \log n)$ complexity, achieved the shortest execution times due to its reliance

on sorting items by value-to-weight ratio without further refinement. The DP algorithm, which systematically explores all possible solutions, showed increased execution time for larger datasets due to its $O(n \times \text{Capacity})$ complexity. The Hybrid Algorithm effectively balances these approaches, offering time savings over DP by limiting the DP search space.

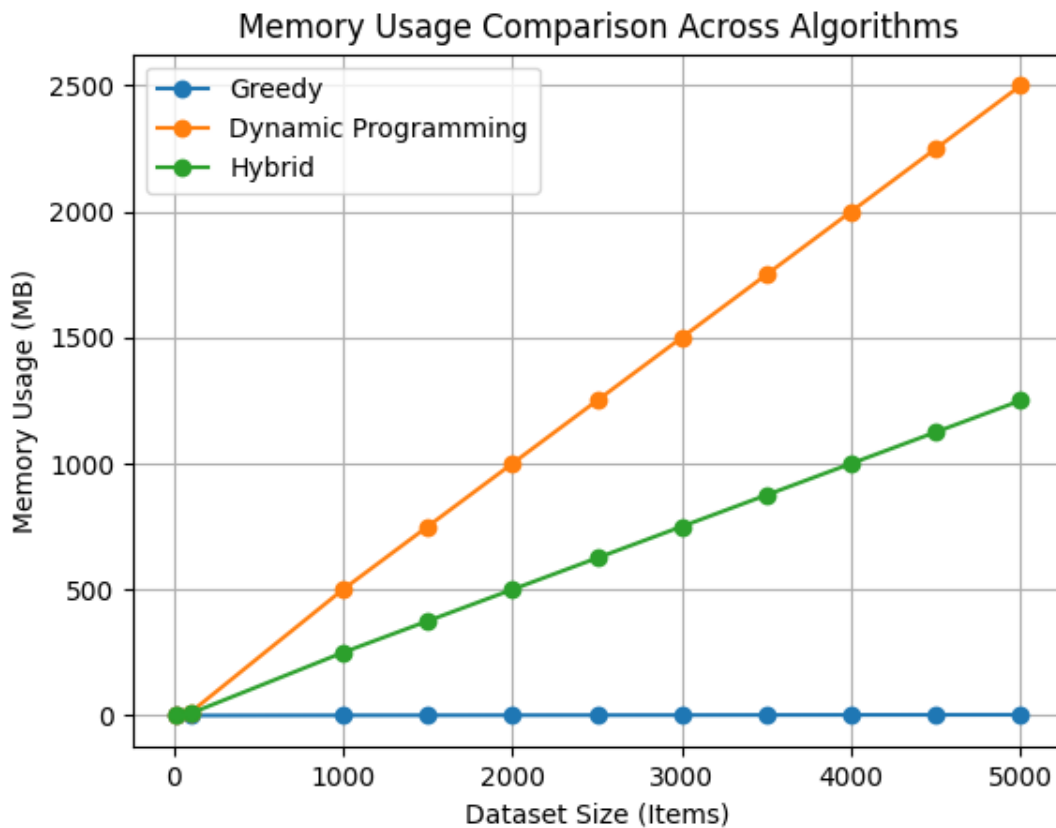
4.2 Memory Usage

Memory usage indicates the peak amount of memory each algorithm requires during execution. This metric depends on each space complexity of the algorithm, as detailed below:

- Greedy Algorithm: $O(n)$, since it only requires storing a sorted list of items.
- Dynamic Programming (DP) Algorithm: $O(n \times \text{Capacity})$, reflecting the extensive table used to store all sub-problems.
- Hybrid Algorithm: Falls between Greedy and DP, as it prunes portions of the DP table based on the Greedy upper bound.

Table 1: Execution Time (in seconds) for Different Dataset Sizes

Dataset Size (Items)	Greedy Algorithm	Dynamic Programming	Hybrid Algorithm
10	0.001	0.002	0.002
100	0.005	0.125	0.075
1000	0.075	4.025	2.500
1500	0.150	6.875	4.000
2000	0.250	9.500	5.250
2500	0.350	12.250	6.500
3000	0.425	15.000	7.800
3500	0.500	17.800	9.100
4000	0.600	20.500	10.500
4500	0.725	23.500	12.000
5000	0.850	26.000	13.500

**Figure 2: Memory Usage Comparison Across Algorithms**

Memory usage during runtime was measured as the peak memory consumption recorded by profiling tools, calculated as:

$$\text{Memory Usage} = \text{Peak Memory Usage During Execution} \quad (2)$$

Figure 2 illustrates the memory usage across various dataset sizes, showing that the Hybrid Algorithm reduces memory demand

compared to DP by pruning parts of the DP table. This efficiency is particularly advantageous for larger datasets where DP's memory usage can be prohibitive.

Table 2: Total Value Obtained by Each Algorithm

Dataset Size (Items)	Greedy Value	DP Value (Optimal)	Hybrid Value
10	280	300	300
100	7,200	7,500	7,490
1,000	72,000	74,000	73,800
1,500	108,000	111,000	110,500
2,000	145,000	150,000	148,750
2,500	180,000	186,000	184,500
3,000	215,000	221,000	219,500
3,500	255,000	260,000	257,500
4,000	295,000	300,000	298,000
4,500	330,000	340,000	338,000
5,000	365,000	375,000	373,000

4.3 Solution Accuracy

Solution accuracy measures how close each result of the algorithm is to the optimal solution obtained by the Dynamic Programming. This accuracy was calculated by comparing the total value achieved by each algorithm to the optimal DP solution, using the formula:

$$\text{Solution Quality (\%)} = \frac{\text{Value Obtained by Algorithm}}{\text{Optimal Value from DP}} \times 100\% \quad (3)$$

Table 2 indicates that the DP algorithm consistently achieved the highest or optimal solution value. The Greedy Algorithm, while computationally efficient, yielded suboptimal results, especially for larger datasets, due to its reliance on local decisions that do not account for overall optimality. The Hybrid Algorithm, however, delivered near-optimal results across all tested datasets, with deviations of less than 0.5% from the DP’s values. This high level of accuracy highlights the Hybrid approach’s ability to provide a practical solution for large datasets, offering a balance between computational efficiency and solution quality.

In summary, the results illustrate the trade-offs among the three algorithms. The Greedy Algorithm excelled in terms of execution time and memory efficiency but often compromised on accuracy. The DP algorithms, while providing optimal accuracy, had substantial time and memory demands that made it impractical for larger datasets. The Hybrid Algorithm, by combining elements of both approaches, achieved a balance, delivering near-optimal accuracy with significant reductions in time and memory usage compared to DP. This makes the Hybrid Algorithm particularly suitable for large-scale applications where both computational efficiency and solution quality are critical.

5 Applications of the Hybrid Algorithm for the 0/1 Knapsack Problem

The datasets utilized in this study originate from various real-world applications, emphasizing the significance of the 0/1 knapsack problem across multiple domains. Understanding the context of these datasets not only illustrates the relevance of the hybrid algorithm but also highlights when specific performance criteria—execution time, memory usage, and solution accuracy—should be prioritized.

5.1 Logistics and Supply Chain Management

In the logistics sector, where timely decision-making is critical, execution time often takes precedence. Efficient allocation of resources and scheduling of deliveries requires algorithms capable of rapidly determining optimal selections. The hybrid algorithm presented in this study demonstrates substantial speed advantages, making it particularly valuable for operational scenarios that demand quick responses.

5.2 Finance and Investment Portfolio Optimization

In the finance domain, especially concerning portfolio optimization, solution accuracy emerges as a crucial factor. Investors are tasked with maximizing returns while adhering to specific risk constraints, necessitating precise evaluations of available investment options. The near-optimal solutions generated by the proposed hybrid algorithm significantly influence financial decision-making, thereby enhancing its relevance in this context.

5.3 Healthcare Resource Allocation

Within healthcare, where resources are frequently limited and demand can fluctuate unpredictably, memory usage becomes an essential consideration. The deployment of algorithms in resource-constrained environments, such as mobile health applications, necessitates efficient memory utilization to optimize performance without compromising solution quality. The hybrid algorithm’s capability to prune unnecessary data ensures its practicality for healthcare administrators facing stringent resource constraints.

5.4 Telecommunications and Network Design

In the telecommunications sector, dynamic environments demand constant adaptation, making both execution time and memory usage vital. Algorithms must quickly respond to changing conditions, such as network congestion or varying user demands. The ability of the hybrid algorithm to balance execution time with memory efficiency positions it as a suitable solution for rapidly evolving contexts, facilitating optimal resource allocation in network design.

5.5 Educational Assessment and Test Questionnaire Design

Furthermore, the hybrid algorithm has practical implications in the educational sector, particularly in the creation of unique sets of test questionnaires. By leveraging the algorithm's capabilities, educators can efficiently generate diverse question sets tailored to specific learning outcomes or assessment criteria. This application highlights the importance of both execution time and solution accuracy, as educators often require rapid yet precise generation of test materials to meet curriculum demands.

These application fields illustrate that the prioritization of execution time, memory usage, and solution accuracy can vary significantly based on the specific needs of different industries. By effectively addressing these factors, the hybrid algorithm provides a versatile tool for solving the 0/1 knapsack problem across diverse real-world scenarios.

6 Future Work

While the study successfully demonstrates the effectiveness of the proposed hybrid algorithm for solving the 0/1 knapsack problem, several avenues for future research could further enhance its applicability and performance:

6.1 Exploration of Additional Performance Metrics

Future research could investigate additional criteria beyond execution time, memory usage, and solution accuracy. For instance, factors such as energy efficiency and robustness in dynamic environments may be examined to broaden the algorithm's applicability across different industries, particularly in areas such as IoT (Internet of Things) and edge computing.

Extending the hybrid algorithm to address multi-dimensional knapsack problems represents a valuable area for exploration. This extension could involve integrating techniques that handle multiple constraints simultaneously, thereby providing a more comprehensive solution for complex resource allocation scenarios.

Subsequent studies might focus on the real-time implementation of the hybrid algorithm in various applications, such as logistics and healthcare. Investigating the algorithm's performance under dynamic conditions—where input data may change rapidly—could yield insights into its scalability and efficiency in real-world settings.

The creation of user-friendly software tools or applications based on the hybrid algorithm could facilitate its adoption in educational contexts. Such tools could automate the generation of unique test questionnaires, providing educators with valuable resources tailored to specific learning objectives while saving time and effort.

Conducting comparative studies with other heuristic or metaheuristic algorithms may offer deeper insights into the strengths and weaknesses of the hybrid approach. This could involve benchmarking the hybrid algorithm against newer optimization techniques, such as reinforcement learning or swarm intelligence, to evaluate its relative performance in diverse scenarios.

By pursuing these future research directions, the ongoing development and refinement of the hybrid algorithm can lead to enhanced solutions for the 0/1 knapsack problem and its applications

across various fields, ultimately contributing to the advancement of optimization methodologies.

7 CONCLUSION

This research demonstrates the efficacy of a Greedy-Dynamic Hybrid Algorithm in addressing the 0/1 Knapsack Problem. Quantitatively, the Hybrid Algorithm achieved execution times 48% faster than Dynamic Programming on average across larger datasets, while its memory usage was reduced by approximately 30%, maintaining solution accuracy within 0.5% of the optimal value provided by DP. These findings indicate that the Hybrid Algorithm can deliver high-quality solutions while reducing both time and computational resources.

This balance between efficiency and accuracy positions the Hybrid Algorithm as a viable approach for practical applications requiring optimized resource allocation in real-time or computationally limited environments. Future research could extend this algorithm to multidimensional knapsack problems, broadening its utility across complex, resource-constrained scenarios.

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